

torch.signal.windows.nuttall#

<https://pytorch.org/docs/stable/generated/torch.signal.windows.nuttall.html>

Description:

Computes the minimum 4-term Blackman-Harris window according to Nuttall. where $z_n = 2\pi n M$ $z_n = \frac{2 \pi n}{M}$ $z_n = M 2\pi n$. The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and sym is True. M (int) – the length of the window. In other words, the number of points of the returned window. sym (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True.

Function Signature

```
torch.signal.windows.nuttall(M, *, sym=True, dtype=None,
layout=torch.strided, device=None, requires_grad=False)
[source]#
```

Parameters

Keyword Arguments

Return type

Parameters

Keyword

`sym` (bool, optional) – If False, returns a periodic window suitable for use in spectral analysis. If True, returns a symmetric window suitable for use in filter design. Default: True. `dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if None, uses a global default (see `torch.set_default_dtype()`). `layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`. `device` (torch.device, optional) – the desired device of returned tensor. Default: if None, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types. `requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default

Returns:

Tensor

Code Examples

- A. Nuttall, "Some windows with very good sidelobe behavior,"
IEEE Transactions on Acoustics, Speech, and Signal Processing,
vol. 29, no. 1, pp. 84-91,
Feb 1981. <https://doi.org/10.1109/TASSP.1981.1163506>
- Heinzel G. et al., "Spectrum and spectral density estimation
by the Discrete Fourier transform (DFT),
including a comprehensive list of window functions and some
new flat-top windows",
February 15, 2002 https://holometer.fnal.gov/GH_FFT.pdf

```
>>> # Generates a symmetric Nuttall window.  
>>> torch.signal.windows.general_hamming(5, sym=True)  
tensor([3.6280e-04, 2.2698e-01, 1.0000e+00, 2.2698e-01, 3.6280e-  
04])  
  
>>> # Generates a periodic Nuttall window.  
>>> torch.signal.windows.general_hamming(5, sym=False)  
tensor([3.6280e-04, 1.1052e-01, 7.9826e-01, 7.9826e-01, 1.1052e-  
01])
```

Detailed Documentation

Documentation

Computes the minimum 4-term Blackman-Harris window according to Nuttall.

where $z_n = 2\pi n M$
$$z_n = \frac{2\pi n}{M}$$

The window is normalized to 1 (maximum value is 1). However, the 1 doesn't appear if M is even and `sym` is `True`.

`M` (int) – the length of the window. In other words, the number of points of the returned window.

`sym` (bool, optional) – If `False`, returns a periodic window suitable for use in spectral analysis. If `True`, returns a symmetric window suitable for use in filter design. Default: `True`.

`dtype` (torch.dtype, optional) – the desired data type of returned tensor. Default: if `None`, uses a global default (see `torch.set_default_dtype()`).

`layout` (torch.layout, optional) – the desired layout of returned Tensor. Default: `torch.strided`.

`device` (torch.device, optional) – the desired device of returned tensor. Default: if `None`, uses the current device for the default tensor type (see `torch.set_default_device()`). device will be the CPU for CPU tensor types and the current CUDA device for CUDA tensor types.

`requires_grad` (bool, optional) – If autograd should record operations on the returned tensor. Default: `False`.

